

In practice, P_e is experimentally obtained from transmitting and detecting a lot of bits and counting the number of bits that is detected incorrectly. Subsequently, one can approximate P_e by taking the ratio of the number of bits that is detected incorrectly to the total number of bits transmitted. This is why P_e is also commonly referred to as bit error ratio (BER). A BER of 10^{-6} means that on average, 1 out of 1000000 bits transmitted will be incorrectly detected.

Now, the BER or probability of detection error P_e is given by

$$\begin{aligned} P_e &= P(0|1) \\ &= \int_0^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-(y+A)^2/2\sigma^2} dy \\ &= \int_A^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-z^2/2\sigma^2} dz \quad (z = A + y) \end{aligned}$$

Such a probability is shown in the following figure. Graphically, the BER is the area of the ‘tail’ of a Gaussian probability distribution.

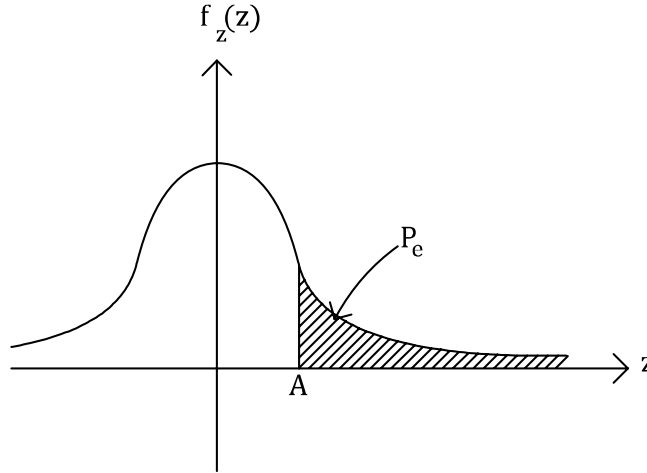


Figure 8.7. Probability of detection error, or bit error ratio (BER), can be graphically represented as the area until the tail of a Gaussian distribution.

This integral happens so often in engineering and physics so that we have a special function for evaluating such integral. The **complementary error function** $\text{erfc}(x)$ is defined as

$$\text{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-z^2} dz.$$

In this case,

$$\begin{aligned} P_e &= \int_A^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-z^2/2\sigma^2} dz \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{A/\sqrt{2}\sigma}^{\infty} e^{-v^2} \sqrt{2}\sigma dv \quad (\text{let } v = z / \sqrt{2}\sigma, \quad dv = dz / \sqrt{2}\sigma) \\ &= \frac{1}{\sqrt{\pi}} \int_{A/\sqrt{2}\sigma}^{\infty} e^{-v^2} dv = \frac{1}{2} \text{erfc}\left(\frac{A}{\sqrt{2}\sigma}\right) \end{aligned}$$

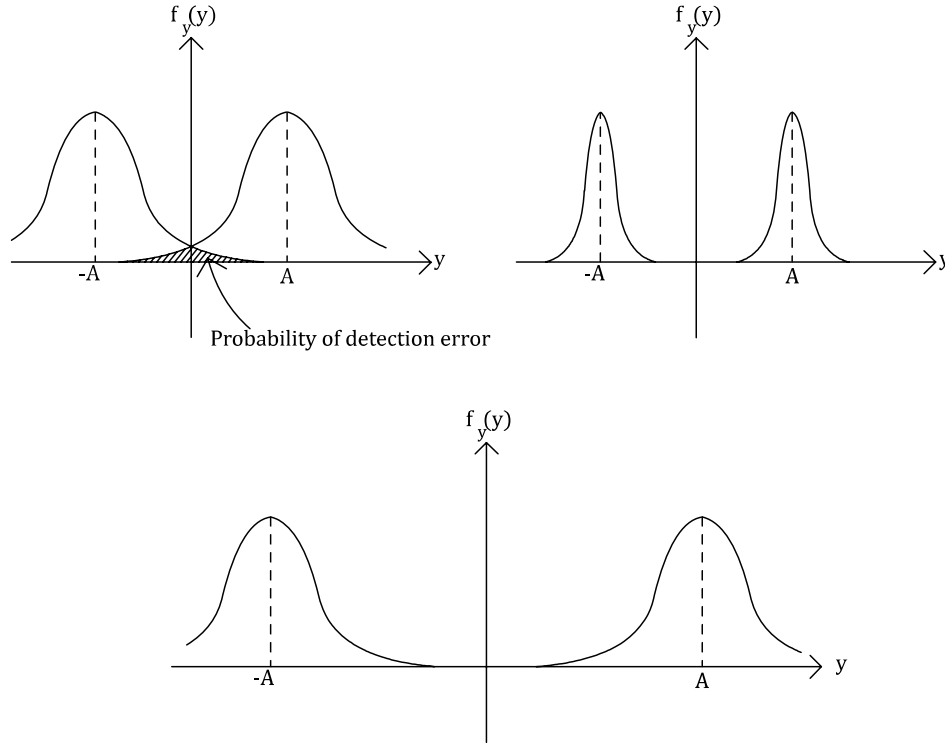


Figure 8.8. Probability of detection error, or BER, and its dependence on signal amplitude and noise standard deviation. BER generally decrease when signal amplitude is increased and/or noise standard deviation is reduced. More precisely, BER decrease with increasing signal-to-noise ratio (SNR), defined as the ratio of the signal power to noise power.

Note that the BER depends on the signal amplitude A and the standard deviation of the noise σ . To reduce BER, we should increase the signal amplitude and/or reduce the noise standard deviation. This is shown in Figure 8.8. More generally, we can consider σ^2 as **noise power** and define the signal-to-noise ratio (SNR)

$$\text{SNR} = \frac{\text{Signal power}}{\text{Noise power}} = \frac{A^2}{\sigma^2}.$$

In this case, the BER is given by

$$P_e = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{\text{SNR}}{2}} \right).$$

For voice and video, the general BER requirement is about 10^{-3} . For data transmission, we typically need a BER to be $< 10^{-15}$ i.e. 1 detection error in 1000000 G bits transmitted ~! The BER requirement and the magnitude of noise power at the receiver (which we cannot control) set a lower limit on the transmitted signal power. Another important to note is that in practice, all commercial communication systems need to meet the BER requirements before it can be used. Therefore, when we compare different service providers with different speed rates, one thing that we always ignore is that all systems need to meet the BER requirement before it can compete with others in terms of speed.

A plot of the BER vs. SNR (in dB) is shown in Figure 8.9. From the figure, it can be shown that if the SNR is doubled (from 20 dB to 23 dB), P_e reduces by more than 10^5 . Therefore, a small change in SNR can result in a huge change in the BER hence performance of a communication system.

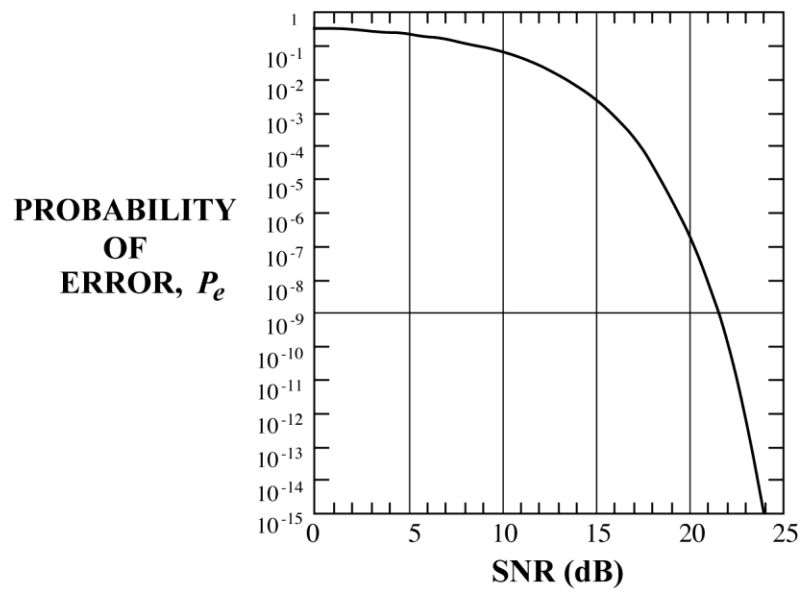


Figure 8.9. Probability of error for thermal-noise-limited systems.

For a more general pulse shape $p(t)$, the signal power should be $S = \frac{1}{T} \int_0^T p^2(t) dt$ instead of A^2 .

The following is the conditional pdfs of a 4-PAM system and illustrate the source of detection errors.

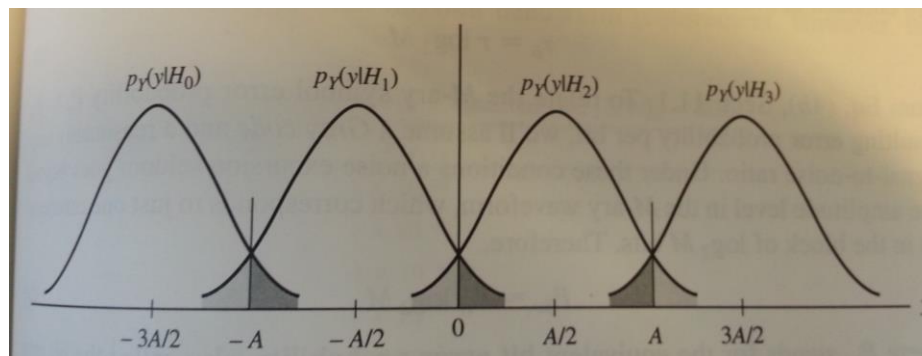


Figure 8.10. Conditional pdfs of a 4-PAM system. The shaded areas represent error detection probabilities.

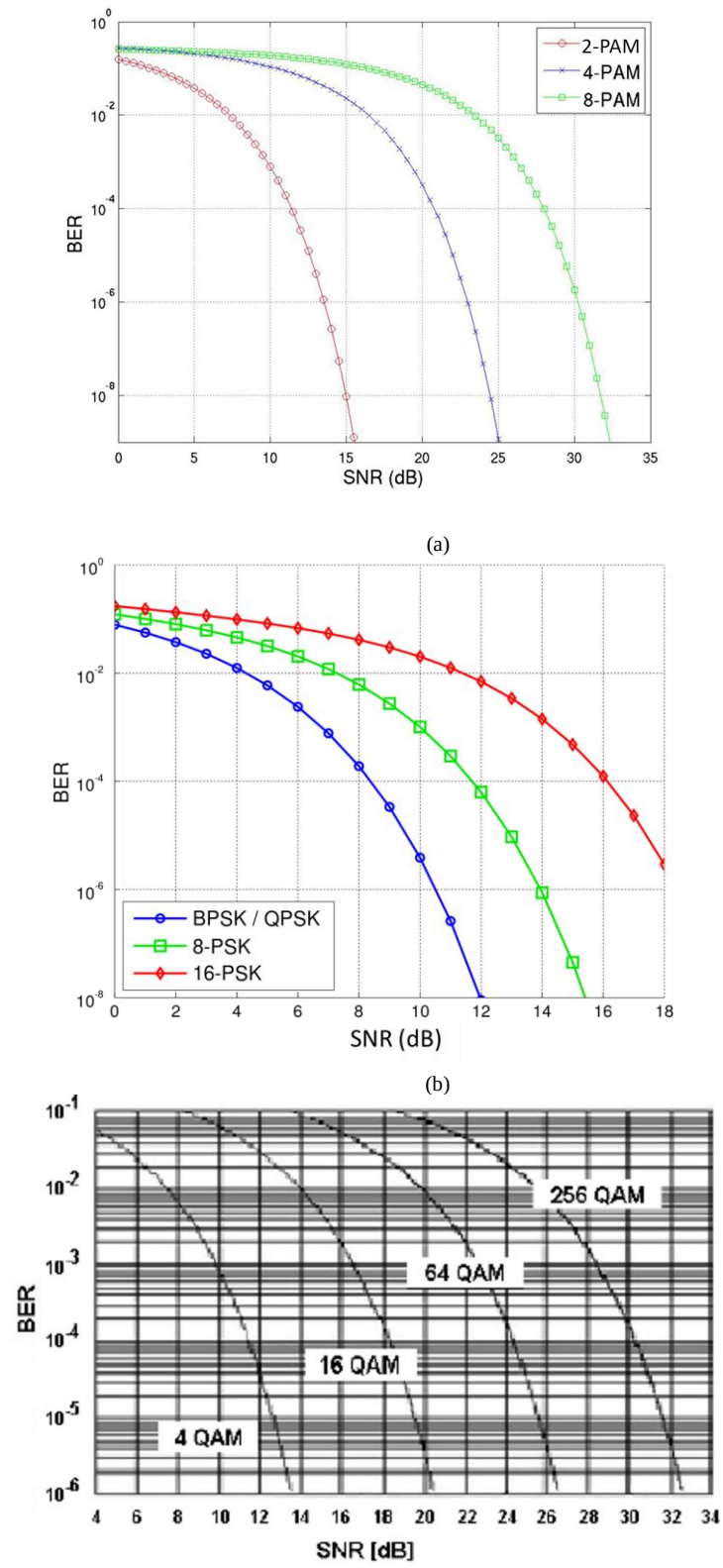


Figure 8.11. Probability of symbol error for (a) M-PAM systems; (b) M-PSK systems and (c) M-QAM systems

As we can see, low enough BER requires a minimum SNR for different modulation formats. In practice, one may not do much about the noise power. On the other hand, signal power arriving at the receiver do vary because of various reasons, and one important reason is signal attenuation along transmission, which is true for both copper wire, wireless and fiber optic channel. Consequently, minimum SNR requirement translate into a maximal transmission distance allowed.

Transmission Medium	Frequency	Loss dB/km
Open-wire pair (0.3 cm diameter)	1 kHz	0.05
Twisted-wire pair (16 gauge)	10 kHz	2
	100 kHz	3
	300 kHz	6
	1 MHz	2
Coaxial cable (1 cm diameter)	100 kHz	1
	3 MHz	4
	100 MHz	1.5
Coaxial cable (15 cm diameter)	10 GHz	5
Rectangular waveguide (5 × 2.5 cm)	100 GHz	1.5
Helical waveguide (5 cm diameter)	3.6 × 10 ¹⁴ Hz	2.5
Fiber-optic cable	2.4 × 10 ¹⁴ Hz	0.5
	1.8 × 10 ¹⁴ Hz	0.2

Figure 8.12. Typical values of transmission loss.

Example 8.1

Consider a Coaxial cable system (15 cm diameter) with transmitted signal power of 1 W and noise power 0.001W. If we are going to transmit PAM-2 i.e. polar binary signals at carrier frequency around 100 MHz with a BER requirement of 10^{-4} , how long can the wire be?

Answer: From Fig. 8.11(a), required SNR for PAM-2 for 10^{-4} BER is around 12 dB. Since $1W/0.001W = 30dB$ and the attenuation of the coaxial cable at carrier frequency 100MHz is 1.5 dB/km, the maximum length allowed will be $(30-12)/1.5 =$ around 12 km.

Example 8.2

Consider a twisted-wire pair system with transmitted signal power of 1 W and noise power 0.001W. If we are going to transmit 8-PSK signals at carrier frequency around 10 kHz with a BER requirement of 10^{-4} , how long can the wire be?

Answer: From Fig. 8.11(b), required SNR for 8-PSK for 10^{-4} BER is around 11.8 dB. Since $1W/0.001W = 30dB$ and the attenuation of twisted-wire pair at 10kHz is 2 dB/km, the maximum length allowed will be $(30-11.8)/2 =$ around 9.1 km.

Example 8.3

Consider a 5-km twisted-wire pair system with transmitted signal power of 1 W and noise power 0.001W. If we are going to transmit 16-QAM signals with a BER requirement of 10^{-4} , what can the carrier frequency be?

Answer: First, $1W/0.001W = 30dB$. From Fig. 8.11(c), required SNR for 16-QAM for 10^{-4} BER is around 18.2 dB. Now, as $(30-18.2)/5=2.36$ dB/km, that means the carrier frequency of 100 and 300 kHz is not suitable while 10 kHz is suitable.

Example 8.4

Consider a twisted-wire pair system with distance 3 km, transmitted signal power of 1 W and noise power 0.001W. If we are going to transmit signals at carrier frequency around 10 kHz with a BER requirement of 10^{-4} , can we transmit 4-QAM, 16-QAM or even 64-QAM? Explain your answer.

Answer: Since $1W/0.001W = 30\text{dB}$ and the attenuation of twisted-wire pair at 10kHz is 2 dB/km, the total attenuation is $2 \times 3 = 6$ dB and hence the received SNR will be $30 - 6 = 24$ dB. Now, from Fig. 8.11(c), the SNR required for $\text{BER} = 10^{-4}$ is 11.5, 18.2 and 24.2 dB respectively. Since the received SNR is 24 dB, we can conclude that the system can support 4-QAM, 16-QAM but not 64-QAM signals. In practice, one will transmit 16-QAM signals to maximize the overall bit rate.